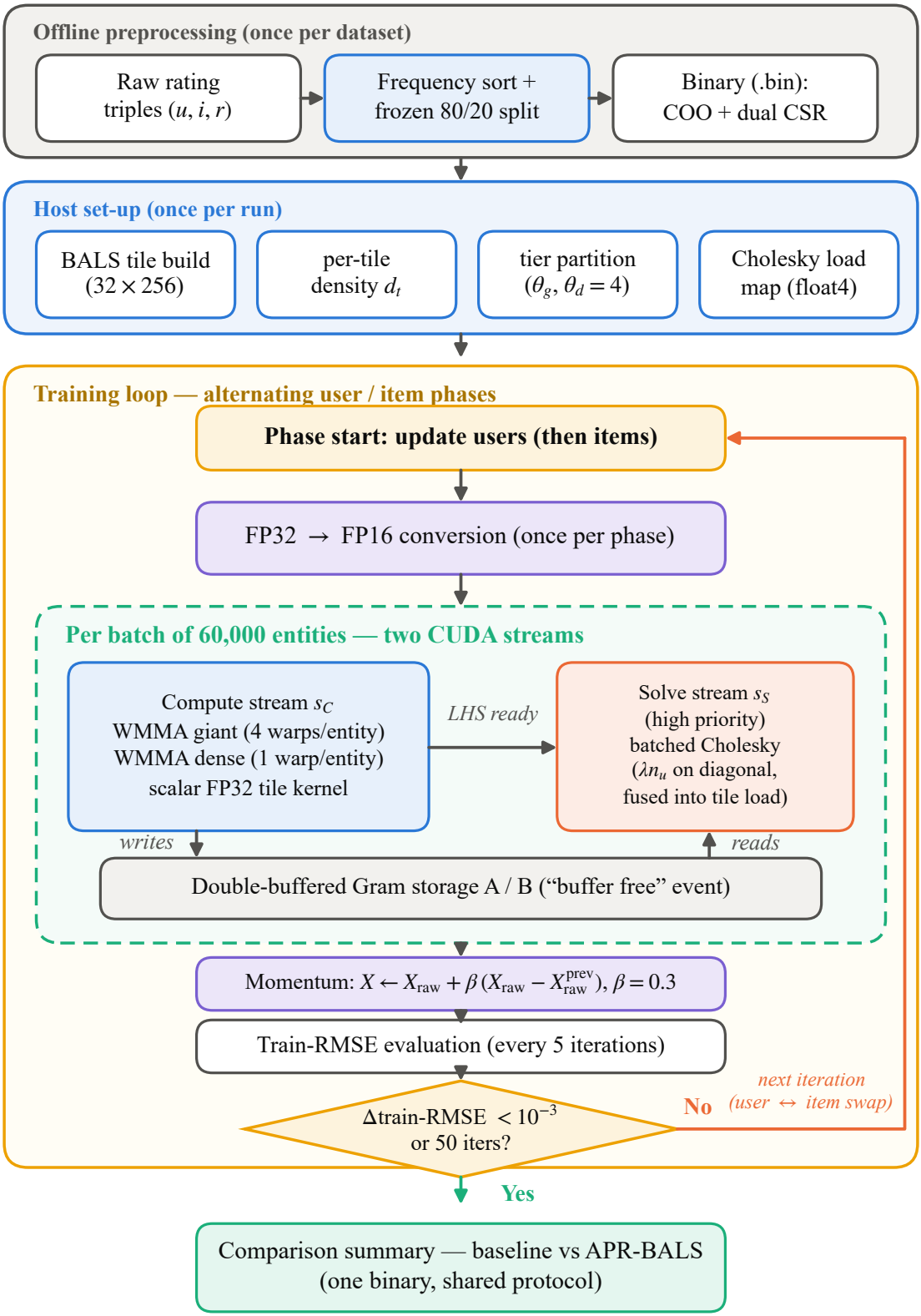

Offline preprocessing (once per dataset)
Raw rating triples (u, i, r)
Frequency sort + frozen 80/20 split
Binary (.bin): COO + dual CSR
Host set-up (once per run)
BALS tile build (32×256)
per-tile density d_t
tier partition $(\theta_g, \theta_d = 4)$
Cholesky load map (float4)
Training loop — alternating user / item phases
Phase start: update users (then items)
FP32 → FP16 conversion (once per phase)
Per batch of 60,000 entities — two CUDA streams
Compute stream s_C
WMMA giant (4 warps/entity)
WMMA dense (1 warp/entity)
scalar FP32 tile kernel
LHS ready
Solve stream s_S
(high priority)
batched Cholesky
(λn_u on diagonal,
fused into tile load)
writes
reads
Double-buffered Gram storage A / B ("buffer free" event)
Momentum: $X \leftarrow X_{\text{raw}} + \beta\,(X_{\text{raw}} - X_{\text{raw}}^{\text{prev}})$, $\beta = 0.3$
Train-RMSE evaluation (every 5 iterations)
Δtrain-RMSE $< 10^{-3}$ or 50 iters?
No
next iteration
(user ↔ item swap)
Yes
Comparison summary — baseline vs APR-BALS
(one binary, shared protocol)